Federica Domecq Lacroze

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EDUCATION

Wake Forest University — Winston-Salem, NC
M.S. Statistics — Anticipated May 2028

Pomona College — Claremont, CA
B.A. Mathematics & Computer Science, Magna Cum Laude, GPA: 3.978/4.00 — May 2026
Honours: Phi Beta Kappa, Sigma Xi, Bruce Jay Levy Senior Prize in Mathematics, Distinction in Mathematics Senior Exercise, Bernard Charnwut Chan Leadership & Service Award, Sheri Sangji Memorial Award
Relevant Coursework: Statistical Theory, Computational Statistics, Statistical Learning, Machine Learning, Time Series, Linear Models, Probability, Real Analysis I–II

TECHNICAL SKILLS

Programming: Python (pandas, NumPy, scikit-learn, matplotlib/seaborn), R (tidyverse, ggplot2), SQL, JavaScript/D3
Tools: Git, LaTeX, Excel, HPC/Parallel Computing, LLM-assisted coding & documentation
Languages: English (Fluent), Spanish (Fluent), French (Intermediate), Portuguese (Intermediate)

RESEARCH EXPERIENCE

Wake Forest University — Winston-Salem, NC
Research Assistant – AI Tools for Experiential Learning in Statistics — Sep. 2026 – Present
- Develop AI-powered tools for practising statistical consulting skills, from client communication to analysis planning.
- Contribute to experimental design, data collection, and analysis evaluating the tools' effectiveness.

Pomona College — Claremont, CA
Kenneth Cooke Research Fellow – "On Par: Leveraging GPR for Perfect Golf, on Average" — May 2025 – Present
- Lead independent research, begun as a summer fellowship, applying spatial data analysis and machine learning to optimise player-specific golf strategy, presenting at eCOTS 2026 and winning first place in the American Statistical Association Undergraduate Statistics Research Project Competition.
- Built a Python/QGIS spatial pipeline and Gaussian Process Regression models (GPyTorch) to estimate continuous scoring-risk surfaces across 280+ course locations, fusing simulated and observed data through a multi-fidelity framework.
- Ran parallel Monte Carlo simulations on HPC clusters to optimise club and aim-point choice for expected strokes and birdie probability.
- Preparing a manuscript for imminent submission to the *Journal of Statistics and Data Science in Sports*, covering strategy optimisation mechanism implementing multi-fidelity GPR and including sensitivity analyses.

Senior Thesis in Mathematics – Distinction — Sep. 2025 – May 2026
- Developed combinatorial and topological proofs extending Arrow's Impossibility Theorem.
- Analysed electoral-system and democratic-freedom data across 170+ countries.

Summer Research Assistant — May 2024 – Aug. 2024
- Modelled gene-expression trajectories in R and developed bootstrap bias corrections.
- Parallelised large-scale simulations across HPC clusters to accelerate model evaluation.

TEACHING EXPERIENCE

Wake Forest University — Winston-Salem, NC
Graduate Teaching Assistant, Department of Statistical Sciences — Aug. 2026 – Present
- Grade for two undergraduate statistics courses and provide weekly in-class instructional support.
- Hold 6 hour-long one-to-one tutoring sessions weekly, booked by students across statistics and mathematics courses.

Pomona College — Claremont, CA
Statistics, Data Science & Probability Teaching Assistant — Jan. 2024 – May 2026
- Assisted instruction across Probability (3×), Introductory Statistics (2×), and Data Science (1×), teaching probability, inference, prediction, data wrangling, and reproducible analysis in R/tidyverse.
- Held 4–10 hours of office hours weekly in 2-hour sessions of 3–20 students, mentoring 150+ students on theory and core concepts as well as R/tidyverse data pipelines, personalised code reviews, and environment setups (RStudio/Positron).

LEADERSHIP & ATHLETICS

Pomona-Pitzer Women's Varsity Golf — 2022 – 2026
Two-Year Team Captain
3× NCAA National Championship appearances, 2 individual collegiate wins, 2× Team Conference Champion
3× All-Conference, 2× All-Region, 4× Women's Golf Coaches Association All-American Scholar

Association for Women in Mathematics — Pomona College
Co-President — 2023 – 2025
Math Department Liaison — 2025 – 2026